Feasibility of AI-Based Conversational Interviews in Health Research: Evaluating a Large Language Model for Semi-Structured Data Collection

Meirong Liu, Ph.D.[1], Janet Currie, Ph.D.[3], Lu He, Ph.D.[2], Jae Eun Chung, Ph.D.[1], Rahual Rai[1], Aaryan Panthi[1], Bibesh Sedhain[1], Jiang Li, Ph.D.[1], Olivia Kasule[3], Chloe Riche[3], Kyaus Washington, MSW[1]
[1]Howard University, District of Columbia, United States; [2]University of Wisconsin-Milwaukee, Milwaukee, Wisconsin, United States; [3] Princeton University, Princeton, New Jersey, United States

Abstract
The present study examines the feasibility and acceptability of using a large language model (LLM)–enabled interview system to conduct qualitative interviews with caregivers of children with asthma living in subsidized housing in Washington, DC. This research is part of a broader community-based participatory research (CBPR) project that emphasizes equitable partnerships, mutual learning, and actionable outcomes. Building on insights from previous human-led interviews and ongoing community collaboration, the study designed and developed an AI-assisted interview tool that is accessible, literacy-adapted, and uses culturally sensitive and empathetic language.

Introduction
Conducting qualitative health research with marginalized populations, such as caregivers of children with asthma in subsidized housing, poses persistent challenges related to trust, disclosure, and resource constraints. In addition, traditional qualitative interview methods are prone to social desirability bias, where participants present responses that are more socially acceptable than their true opinions or behaviors[1]. Recent advancements in Large Language Models (LLMs) offer potential solutions to this challenge by simulating human-like conversations[2]. By significantly reducing the cost associated with human-led interviews, LLMs could enable scalable and detailed conversations[3] and address the limitations of traditional data collection methods. This study aimed to evaluate the usability and feasibility of an AI-based interview tool powered by a Large Language Model (LLM) as an alternative for conducting semi-structured interviews with marginalized populations in the context of health.

Methods
Community engagement was central to the design and development of the LLM–powered interview system. Interview protocols were reviewed for cultural and linguistic appropriateness, pilot tested with participants from prior human-led interviews and refined based on feedback from community partners. Input from human-led interviews and community partners guided the system's design, including considerations such as literacy level, accessibility, and user experience.

Two large language models—Google Gemini 2.0 Flash and OpenAI o3-mini—were evaluated as potential backbones for the interview system. Both models were programmed to follow the same semi-structured interview guide used in previous human-led sessions, maintaining consistency in content while enabling a natural conversational flow. The interview questions focused on policy and its implementation to reduce home asthma triggers, resources and support needs, asthma education, and recommendations for equitable and sustainable asthma care networks. The AI-powered interview tool, *Asthma Care Talk*, was developed through an iterative process involving multiple rounds of pilot testing and refinement. In the first phase, research team members and community partners pilot-tested the system, providing feedback on usability, clarity, and tone. A second phase of testing identified additional refinements.

Participants were caregivers of children with asthma living in subsidized housing in Washington, DC. Recruitment occurred through existing community partnerships and prior engagement networks developed through ongoing community-based research initiatives. A total of 29 caregivers participated between July 1 and August 20, 2025. Written informed consent was obtained from all participants before the study began, and participation was voluntary.

Before each session, the research team explained the study's purpose and demonstrated the AI interview system. During every session, two researchers were present—one to provide technical support and the other to observe and record usability issues, participant engagement, and behavioral cues. Participants were invited to an evaluation session to provide feedback on their AI-led interview. They completed a standardized evaluation form assessing their perceptions of the AI interview experience, including ease of use, response quality, and engagement. All sessions were audio recorded for qualitative analysis. Participants received a $50 gift card and an informational flyer with a QR code linking to the asthma video library. The study protocol was approved by the university's Institutional Review Board. Data analysis drew from three main sources: LLM-powered interview transcripts, transcripts from evaluation session discussions, and researcher observation notes. Text logs from AI-led interviews were de-identified, cleaned, and organized. Audio feedback sessions were transcribed verbatim using Otter.ai and verified for accuracy. Observation notes provided contextual information about usability, participant comfort, and communication patterns during sessions. Thematic analysis was conducted following the approach described by *Jowsey et al.* [4]

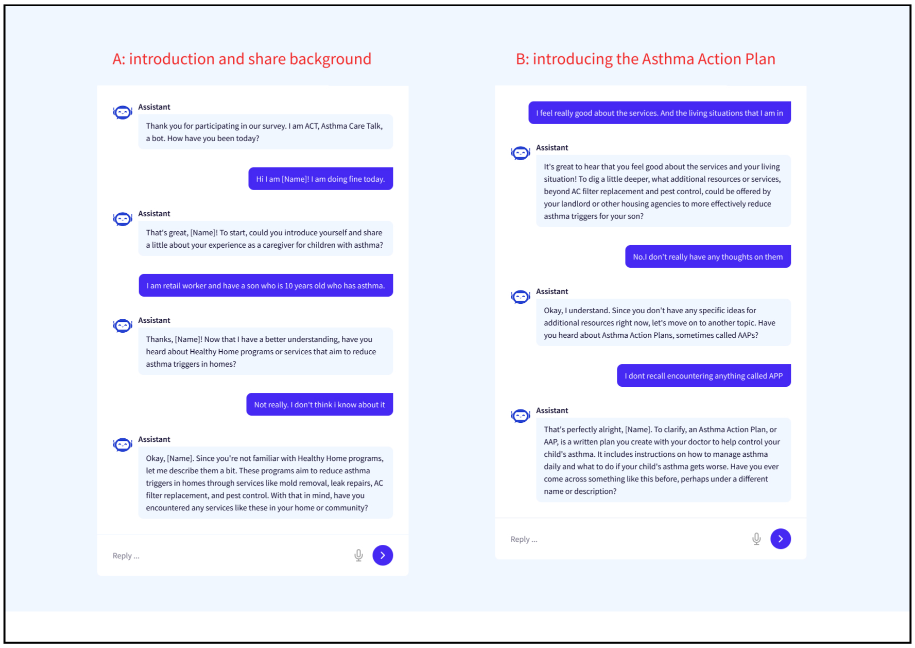

Figure 1. LLM-Powered Interview System Interacting with a Participant

Results

Participants reported that the AI interviewer was a) easy to understand, b) felt comfortable during the LLM-led interviews, descripting the questions as non-invasive and relevant to their lived experiences, with some participants linked their comfort to both the system's empathetic tone and the support of researchers present during the sessions. c) the system was easy to use and intuitive, particularly citing features such as clickable buttons, visual layout, and voice options as key aids to accessibility; d) appreciated its empathetic and nonjudgmental tone. The system successfully elicited detailed responses, demonstrating potential for generating rich qualitative data. However, several usability challenges were identified, including intermittent technical errors, limited contextual adaptability, and differences in participant comfort with digital technology.

Discussion and Conclusions

The findings indicate that AI-driven conversational interview systems are a feasible approach for conducting semi-structured qualitative interviews. The project's community-based participatory research (CBPR) framework was integral to its success. Continued refinement of system design, interaction flow, and ethical safeguards is needed to enhance the reliability and acceptability of LLM-based tools in health and social research.

References

1. Bispo Júnior JP. Social desirability bias in qualitative health research. *Rev Saude Publica.* 2022;56:101.
2. Palmer A, Smith NA, Spirling A. Using proprietary language models in academic research requires explicit justification. *Nat Comput Sci.* 2023 Dec 21;4(1):2–3.
3. Wuttke A, Aßenmacher M, Klamm C, Lang MM, Würschinger Q, Kreuter F. AI conversational interviewing: transforming surveys with LLMs as adaptive interviewers [Internet]. 2024 [cited 2025 Mar 22]. Available from: https://arxiv.org/abs/2410.01824
4. Jowsey T, Deng C, Weller J. General-purpose thematic analysis: a useful qualitative method for anaesthesia research. *BJA Educ.* 2021;21(12):472–8. doi:10.1016/j.bjae.2021.07.006